SATYA BONDA

Principal AI Architect · Agentic AI Architect · Principal AI Engineer — Multi-Agent Orchestration · MCP · RAG · LLM Evaluation & Observability · AI Governance · Azure & AWS

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SUMMARY

Principal AI Engineer and agentic AI architect with 23 years in enterprise technology: 18 leading delivery at IBM, Oracle, Volvo, Trane and Wipro, hands-on AI engineering since 2021 and production agentic systems since 2024. Builds LLM applications that businesses can trust with real work: multi-agent systems on MCP that show their sources, RAG grounded in customer documents, guardrails and governance in code, traced with OpenTelemetry, shipped as containers on Azure and AWS and released through evaluation gates and canaries. Takes a business workflow from first workshop to scalable, auditable production. Each system below links to its code or running app. Claude Certified Architect – Professional; Microsoft Agentic AI Solutions Architect – Expert.

CORE COMPETENCIES

AI solution architecture · agentic AI & multi-agent orchestration · Model Context Protocol · RAG & hybrid search · LLM evaluation · LLM observability & tracing · Docker, Kubernetes & cloud · LLMOps & CI/CD · guardrails, security & AI governance · Azure AI Foundry · Amazon Bedrock · latency & cost control · technical leadership.

SKILLS

Agentic AI & Multi-Agent Orchestration: supervisor–worker, hub-and-spoke and handoff patterns, planner–critic–executor, generator–critic, fan-out/fan-in, ReAct, tool calling, human-in-the-loop, checkpointing, model routing.

Frameworks & Protocols: MCP, FastMCP, A2A, LangGraph, LangChain, OpenAI Agents SDK, Claude Agent SDK, CrewAI, AutoGen, Microsoft Agent Framework, LiteLLM, FastAPI, TypeScript, React, NestJS, Twilio, Deepgram.

Claude Architecture (CCAR-P, applied): agentic solution design, model tiering, context engineering, prompt caching, structured output, MCP; Claude Code.

Microsoft Agentic AI (AB-100 · AI-103 · TITAN · Agentic Hackathon): Azure AI Foundry, Azure AI Search, Azure OpenAI, Copilot Studio, Power Platform, Microsoft 365 Copilot agents, Dynamics 365 F&O, Agent 365 governance.

Generative AI & Retrieval: RAG, GraphRAG, agentic RAG, hybrid search (BM25 + vector), reranking, embeddings, vector databases (pgvector, Pinecone, FAISS, Azure AI Search), prompt engineering, fine-tuning (LoRA, QLoRA).

Machine Learning: linear regression (Ridge, Lasso, Elastic Net), logistic regression, k-nearest neighbors, SVM, decision trees, random forests, gradient boosting (AdaBoost, XGBoost), Naive Bayes; k-means, hierarchical clustering, DBSCAN, PCA; feature engineering, cross-validation, grid search, precision/recall/F1, ROC-AUC; NLP.

Data & Python: Python, SQL, PostgreSQL, pandas, NumPy, scikit-learn, PyTorch Lightning, PEFT; ETL pipelines.

Evaluation & Release Gates: golden datasets and regression testing, offline and online evaluation (nightly on production), LLM-as-a-Judge calibrated to human review, red teaming, drift detection, canary rollouts with rollback.

Observability & Reliability: OpenTelemetry distributed tracing (Application Insights, AWS X-Ray), trace and correlation IDs across agents and tools, structured JSON logs (KQL), CloudWatch alarms (p95 latency, groundedness, 5xx), Prometheus, Grafana, LangSmith, token and cost ledger, retries, fallbacks, circuit breakers, idempotency for distributed systems; trace-based debugging, root-cause analysis.

Containers, Cloud & LLMOps: Docker, Docker Compose, Kubernetes (Deployment, HPA), Azure Container Apps, Amazon ECS Fargate, infrastructure as code (Bicep, AWS CDK, Terraform), GitHub Actions CI/CD with OIDC, SBOM and secret scanning, MLflow, Key Vault, managed identity, Entra ID, Amazon Bedrock, AgentCore, Lambda.

AI Security, Responsible AI & Governance: least-privilege tool access, agent identity and role-based access, prompt-injection defense, PII/PHI redaction, Azure AI Content Safety, OWASP LLM Top 10, EU AI Act, audit trails.

AI Product & Leadership: AI strategy and use-case qualification, AI product management, technical leadership and mentoring, stakeholder communication, ROI.

CERTIFICATIONS

Anthropic Claude Certified Architect – Professional (CCAR-P), 2026 · Microsoft Agentic AI Business Solutions Architect – Expert (AB-100, 2026) · Microsoft Azure AI Apps & Agents Developer Associate (AI-103, 2026) · Microsoft TITAN Frontier Transformation Engineer (2026).

Also: PMP, PMI-RMP, CSM, SAFe POPM, ITIL V3, Azure Solutions Architect Expert, AWS Cloud Practitioner, Oracle GenAI Professional, Oracle ML Associate.

EXPERIENCE

Principal AI Engineer

Independent AI Consultant · Bengaluru, India · Mar 2026 – Present

Forward-deployed, client-facing and remote: builds and runs Python agentic systems in collaboration with client stakeholders, from discovery to production on Azure and AWS cloud, traced with OpenTelemetry behind eval gates.

- **Microsoft Agentic AI Hackathon (Oct 2026, via client Fusion5):** All four Azure AI Foundry builds scored 100/100, among them a multi-tool MCP agent with a PII guardrail and multimodal RAG.
- **AI enablement:** Taught production best practices to EY and PwC cohorts (Aug–Sep 2026, 20+ participants, 4.7/5), and published the runbook listed below.

Autonomous Agentic AI Auditor — Microsoft AI Audit Packs (Claude · MCP · Multi-Agent) · Oct 2026

- **Problem:** A Microsoft partner must prove each AI project followed a disciplined lifecycle, Responsible AI gates included, which means up to 74 audit templates per project, written by hand from scattered documents.
- **Built:** Nine agents in a fixed chain under a deterministic supervisor draft each template only from the project's own documents, quoted word for word. A critic on a different model and a code check of every quotation gate each draft; a gap becomes a client request, never invented text.
- **Outcome:** In its first engagement, an Australian public-sector AI assistant, it drafted all 46 required templates from 17 documents in about four unattended hours, with zero fabricated citations, for under US$10 in model cost.

Flick — Invoice Reconciliation (LangGraph · MCP · Claude) · Apr – May 2026 · Case study

- **Problem:** Finance analysts at a pest-control multinational spent 30–90 minutes proving every line of each supplier invoice against contracts, purchase orders and meter data, by hand.
- **Built:** A LangGraph manager hands each invoice to specialists, each on its own least-privilege MCP server, and a second-vendor critic rejects uncited claims. Confidence routes green to posting, amber to an analyst and red to a dispute a person releases. Correlation IDs trace every call, and idempotency keys make re-runs safe.
- **Outcome:** Analysts ask a Copilot Studio agent in Teams instead of opening invoices. In pilot it reconciles 1,200 invoices a month, 87% right first time, in about 4 minutes against 47 by hand. Each change replays overnight on 200 analyst-labeled invoices before a 5% canary with auto-rollback.

Compass — Live Dynamics 365 Agent (Azure AI Foundry · MCP) · Jun 2026 – Present · Case study

- **Problem:** Staff needed answers on orders, stock and procedures without learning Dynamics 365 screens.
- **Built:** A planner reaches only 39 read-only Dynamics entities and a search of company procedures; figures are computed in code and no answer exceeds the user's role. It serves Teams, web and MCP tool packs with per-client keys, rate limits and a kill switch, and Agent Factory finds agents by hybrid retrieval or admits no match.
- **Outcome:** Staff get cited answers in production. Each answer leaves a trace of its tool calls, retrieval, checks and latency (3,738 so far; p95 2.1 s) in Application Insights. Every night, an 88-case battery (12 of them OWASP LLM Top 10 security attacks) and an LLM judge at 95% agreement with human reviewers re-test production and file a diagnosed GitHub issue on failure. It has shipped 14 releases in 14 weeks, with one-command rollback.

Quote Intelligence — Quote Comparison (Agents SDK · MCP) · Apr 2026 · Case study · Live · Repo

- **Problem:** A homebuilder's consultants spent half a day on each rival quote, and chatbots invent prices.
- **Built:** Six agents work each quote, including parallel readers, an ambiguity flagger and a different-model verifier. No price enters without its page and source text, which code checks; arithmetic runs in exact decimals.
- **Outcome:** A consultant gets a cent-exact comparison they can defend in 6–8 minutes. On a real job it flagged 26 items and withheld the saving until resolved; 63 checks, 26 adversarial, have never let a fabricated figure through. Live and Repo show the public edition, a simpler Claude pipeline with the same human-review stop.

Enterprise AI Playground — Agentic AI Platform, built solo (LiteLLM · LangGraph) · Mar 2026 · Live

- **Problem:** Every team wants frontier models and agents, but enterprises end up with disconnected subscriptions, no single view of spend, no safe path from prompt to production agent and gaps in AI governance.
- **Built:** An agentic platform where teams discover, build, evaluate and operate agents in one governed place. Discover brings together 30 models from 11 providers, 175 agent blueprints, a 7,547-server admin-approved MCP marketplace and 338 skill packs. Build runs agents on LangGraph as multi-agent systems, such as supervisor-and-worker teams and reviewer panels, that pause for human review and resume from checkpoints; any blueprint then exports as an OpenAI Agents SDK, LangGraph, CrewAI or Microsoft Agent Framework project with deploy scripts. One gateway checks identity, role, budget and key on every call, routes it to an Economy, Workhorse or Premium model and writes a ledger row of tokens, latency, cost and trace ID per agent run.
- **Outcome:** It is live on Azure Container Apps (Docker images in ACR, Bicep, Key Vault, managed identity, autoscaling), with pytest and Playwright in CI. In Evaluate, golden sets and a rubric judge lift each agent up a five-level production-readiness ladder, and a nightly canary that detects drift or latency regressions rolls it back.

Further 2026 systems · built in client consulting, hackathons and training workshops

Denial Navigator, Hospital Claim Denials · Oct 2026 · Live

- **Problem:** Hospitals lose revenue when a denied claim goes unworked before its appeal deadline.
- **Built:** Gives every denied remittance line one action, owner and deadline. Rules decide, gradient boosting ranks by expected recovery, Azure OpenAI explains and cites, a person approves. Synthetic-data proof of concept.

P2P Sentinel and AI Full-Stack Agentic Zone · Oct 2026 · Repo · Live

- **Problem:** Teams need a worked reference architecture for taking an agent from plan to production on Azure.
- **Built:** Ships a tested procure-to-pay agent and a public Claude Code runbook. Intake, match and risk agents gather evidence, code decides and a resolution agent explains, with OpenTelemetry observability: a span per step and tokens per case. Runs on Azure Container Apps, with a Kubernetes manifest and golden-set CI gates.

Bedrock Agentic RAG · Oct 2026 · Repo
- **Problem:** A chatbot over company documents will answer confidently even when the sources do not support it.
- **Built:** A self-correcting LangGraph agent on Amazon Bedrock grades what it retrieves, rewrites queries that miss, and has a judge model check each claim against its cited passages. Deployable on ECS Fargate with AWS CDK, with p95 alarms, circuit-breaker rollback and a post-deploy eval gate.

Copilot Prompt Playbook · Oct – Nov 2026 · Repo · Live
- **Problem:** A 74-use-case Microsoft 365 Copilot program needed something every department could actually use.
- **Built:** Gives each department tested prompts, built-in agents and ready-to-build packs: 47 prompts scored 4.9/5 on a five-criterion rubric and 12 declarative agents tested on 123 questions, refusals included.

IQR, SOX 404 Audit-Evidence Validation · Aug 2026 · Repo
- **Problem:** SOX 404 testers must prove every conclusion from workbooks, emails and screenshots, every period.
- **Built:** Returns pass, pass-with-gaps or fail per check, each claim citing the exact cell or email line. A fixed LangGraph pipeline runs a plan an SME has frozen, so the same input gives the same verdict; Python does the numbers, a blinded verifier re-performs each finding, and five evaluation gates guard release (Azure AI Foundry).

Kala Chakra, Vedic Horoscope Studio · Aug 2026 · Repo
- **Problem:** A horoscope is only credible if every planetary position and date is calculated rather than generated.
- **Built:** A deterministic engine computes every position and date; OpenAI Agents SDK specialists narrate them in English, Hindi or Telugu, and uploaded charts are cross-checked. 42 of 42 evaluation checks pass.

DentaScribe, Clinical Scribe for Dentists · May 2026 · Repo · Live
- **Problem:** Dentists must turn each patient visit into a compliant clinical note with correct billing codes.
- **Built:** Five agents turn a visit transcript into a SOAP note with CDT codes for sign-off. Claude writes, a second Claude reviews, and code ensures every claim quotes the transcript and every code comes from a sealed list.

Dragonfruit AI Production OS · Mar 2026 · Case study · Repo
- **Problem:** A YouTube studio needed workflow automation so each pod could carry seven clients instead of four.
- **Built:** A 79-person time audit found coordinators spending 40% of the week on ClickUp admin, so the case study designs sixteen role tools and four role agents: agents propose, seven human gates decide. The repo holds the first, a Producer agent that turns requests into ClickUp tasks behind a human gate (OpenAI Agents SDK, MCP).

Domain LLM Fine-Tuning, EN→NL · Jan – Mar 2026 · Repo · Live
- **Problem:** Off-the-shelf translation models are not tuned for software text such as UI strings and product specs.
- **Built:** Compares full fine-tuning of MarianMT against LoRA adapters on BLOOM-560M (PyTorch Lightning, PEFT), scoring each with BLEU, chrF, TER and COMET on an 84-sentence software-domain test set.

PMO CoPilot · Jan 2026 · Repo · Live
- **Problem:** PMO leads need status reports, risk views, earned value and steering packs on demand, not by hand.
- **Built:** An orchestrator hands each PMO question to one of eleven specialist agents (OpenAI Agents SDK). Earned-value math is deterministic code, ML models predict delays, and a no-key demo mode runs offline.

Dentsi, AI Voice Receptionist · Jan 2026 · Repo · Live
- **Problem:** Small dental practices miss calls outside front-desk hours, when no one is there to answer.
- **Built:** A voice agent answers, identifies the caller, triages symptoms and books, moves or cancels appointments by tool calling across 18 tools over Twilio, Deepgram and ElevenLabs, on NestJS and PostgreSQL.

Agentic CoLearn · Jan 2026 · Repo · Live
- **Problem:** CBSE students in Grades 1–12 need lessons that match their textbooks; generic AI makes things up.
- **Built:** Lessons are generated only from retrieved NCERT chapter text, and an adversarial verifier tries to refute each one before a child sees it. Mastery and revision are pure math the model never touches (Bayesian Knowledge Tracing, 2-PL IRT, FSRS), and a router falls back across five model providers.

Also in 2026
- **Deep Research Agent** · May · Repo: LangGraph company research that asks first and validates before writing.
- **Copilot Studio Agent Client** · Apr · Repo: React, TypeScript client with Entra ID, tool traces, Adaptive Cards.
- **Patient Intelligence Console** · Jan · Repo: FAISS vector search, the agent locked to one patient; synthetic data.

AI Program Manager / Agentic AI Architect

Insight Global · Bengaluru, India · May 2024 – Feb 2026
- **Fortune-100 AI programs:** Provided technical leadership to an 8-person GenAI team across four AI programs for a Credit & Collections organization, from AI strategy and use-case qualification to adoption, with monthly executive communication. Accounts under AI doubled, utilization reached 71% and $220M was collected.
- **SOX 404 agentic co-worker:** Runs one validation plan per control every period and ships no conclusion without cited evidence, behind golden-set evals and SME review. MVP delivered on three controls (open edition: IQR).

2025 · Agentic systems built and published alongside this role

Supply Chain Intelligence · Dec 2025 · Repo · Live
- **Problem:** A supply-chain decision has to weigh risk, demand, suppliers, inventory and logistics at the same time.
- **Built:** An orchestrator (OpenAI Agents SDK) hands each question to one of six specialists, who use live risk signals (Tavily, NewsAPI), deterministic forecasting and supplier calculators, and a Q-learning inventory policy.

Document Anomaly Detection · Nov 2025 · Repo
- **Problem:** Finance back offices must reconcile every invoice against its leasing contract and purchase order.
- **Built:** Agents extract fields, flag mismatched dates, amounts and schedules, each with a confidence, and route doubtful ones to a person. Adds XGBoost training and an AWS path (S3, DynamoDB, CloudWatch).

Healthcare Prior Authorization · Nov 2025 · Repo · Live
- **Problem:** Clinic staff read doctors' notes by hand, look up codes and assemble each prior-auth packet.
- **Built:** A fixed pipeline of LLM steps extracts CPT and ICD-10 codes, which regex validation checks, drafts the request, tracks the payer's response and prepares an appeal when it is denied. Prototype on synthetic patients.

AI Sales Enablement on MCP · Aug 2025 · Repo · Live
- **Problem:** Sales reps need account, deal and call-transcript answers in one place, not across separate tools.
- **Built:** Four Model Context Protocol servers expose 20 tools, with CRM and analytics behind one FastAPI gateway, and a confidence tool labels each action auto-run (0.90+), review (0.70+) or human-required.

Autonomous Email Agent · Aug 2025 · Repo
- **Problem:** An inbox agent is only useful if it never sends anything its owner has not approved.
- **Built:** A LangGraph agent triages each email and drafts replies and meetings, then pauses at an interrupt so the owner can accept, edit or redirect it in Agent Inbox; each correction is written back into agent memory. Golden-dataset evals with an LLM judge run in LangSmith. Extends LangChain's agents-from-scratch.

Also in 2025
- **Multi-Agent Data Analysis** · Dec · Repo: Turns plain-English questions over CSV files into statistics and charts.
- **SDLC Agents** · Jun – Sep · Repo: CrewAI crew of four turns a brief into plan, code, tests and docs (ChromaDB).
- **Agentic Invoice Processing** · Aug · Repo · Live: Nine-agent prototype with a human-review dashboard.
- **Learning Path Generator** · Mar · Repo: Generator–critic pair that builds learning paths from live web and video.
- **Multi-Model Research Agent** · Feb · Repo · Live: Sourced reports, an MCP server and three-provider fallback.
- **Sage Mind** · Feb · Repo · Live: AutoGen tutor; five nested reviewers clear every draft before a learner sees it.

AI Engineer & Project Manager

Wipro · Bengaluru, India · Jul 2021 – Apr 2024
- **Machine learning:** Built supervised and unsupervised models end to end, with feature engineering, cross-validation and grid search. Two reached production: an Azure ML random-forest order classifier (85%+ accuracy) and XGBoost invoice anomaly detection that learns from reviewer feedback (F1 0.72 → 0.91).
- **MLOps:** Built model serving APIs with MLflow tracking, Azure ML pipelines, CI/CD and drift monitoring.
- **Generative AI experiments:** Natural-language-to-SQL (Azure OpenAI, Streamlit) and a conversational RAG assistant that cites uploaded documents (LangChain, FAISS, GPT-4). Repo (published 2025).
- **Copilot Studio & Power Platform:** Built Copilot Studio agents over Dataverse and SharePoint and a knowledge-grounded ServiceNow virtual agent; won AI Hackathon 2024 with GPT-4 brand-voice compliance.
- **Training:** Ran long-form training programs alongside delivery: Bristlecone (ML, 6 months), National Grid (analytics, 12 months) and Navikenz (GenAI, 5 months).
- **Delivery leadership:** Owned the Oracle CPQ/MDM roadmap; MDM modernization saved ~$900K a year and 3 FTE. Directed the EBS and Primavera migration to OCI and built Python data pipelines into Dynamics 365.

EARLIER EXPERIENCE

Carrier (contract) | Project Manager · Feb – May 2024 · Oracle EBS separation, Kinaxis and LeanDNA.
Trane Technologies / Ingersoll Rand | Project Manager & Agile Coach · Jun 2015 – Jul 2021

Over 6 years, delivered the Oracle 19c upgrade across five workstreams, 10 production databases and 8 applications under budget, led a zero-disruption ERP separation through the company split (President Award) and coached teams in Agile.

Volvo | IT Project Manager · 2013 – 2015 · Oracle | Project Management Consultant · 2011 – 2013

Volvo: ITSM processes that lifted service reliability 35%. Oracle: managed a high-tech and oil-and-gas portfolio.

IBM | Project / Operations / Transition / Quality · Jan 2007 – Apr 2011

Ran five helpdesk accounts with 55 direct reports and stood up 12–20 FTE service desks for four clients.

Genpact | Process Leader · Dec 2005 – Jan 2007 · HP | Quality Specialist · Jan 2004 – Nov 2005

EDUCATION

Master of Computer Applications (MCA), Computer Science, Periyar University (2003) · B.Sc. Physics (Hons.), University of Calcutta (1999)